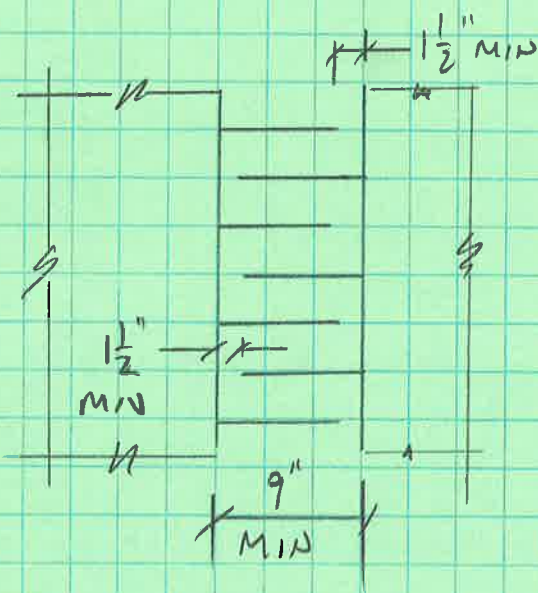




Project	WF-D6			Sheet No.	of	Sheets
S.R.	Made By	Check by	Date	Supv		
	RAB		8/4/17			

WF-D6 JOINT FIT UP - LATERAL DEFLECTIONS

JOINTS MUST BE AT LEAST 9" WIDE WITH $\frac{1}{2}$ " END
CONTR ON BARS



DESIRED JOINT
CONFIGURATION

3 SOURCES OF JOINT VARIATION

1) SWEEP

2) LATERAL DEFLECTION DUE TO PS ON ASYMMETRIC SECTION

3) LATERAL DEFLECTION DUE TO INSTALLATION ON SLOPE



Project				Sheet No.	of	Sheets
S.R.	Made By	Check by	Date	Supv		

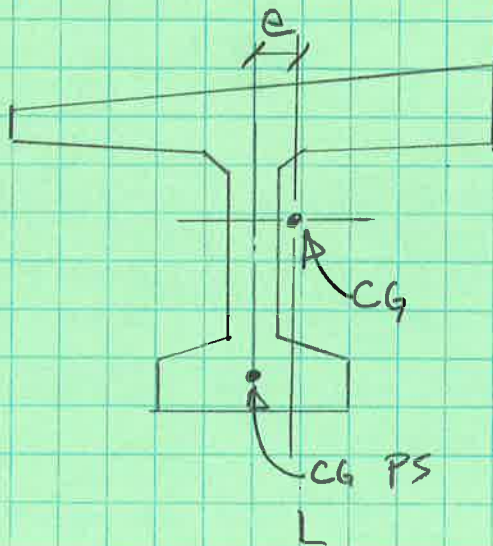
SWEEP

INSDOT SWEEP TOLERANCE IS $\frac{1}{8}" / 10 \text{ ft}$

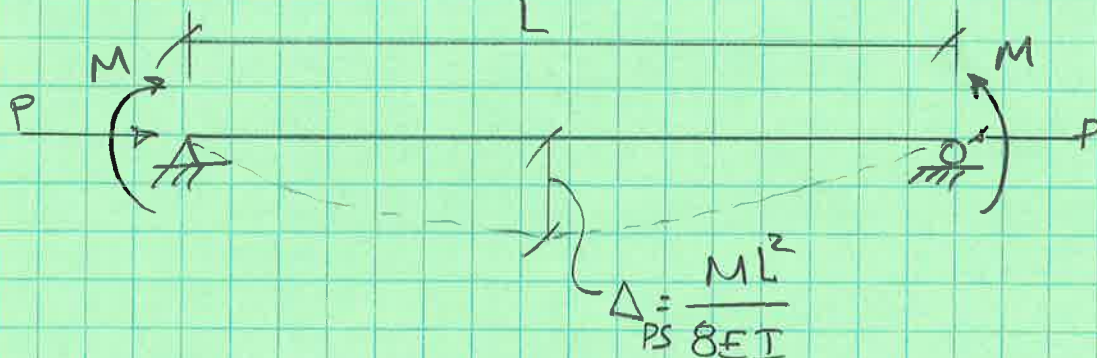
$$\Delta_{\text{SWEEP}} = \left(\frac{1}{8}" \right) \left(\frac{L}{10} \right)$$

ASYMMETRIC SECTION

WHEN CROWN SLOPE IS BUILT INTO TOP FLANGE
THE GIRDER SECTION IS ASYMMETRIC



$$M = Pe$$



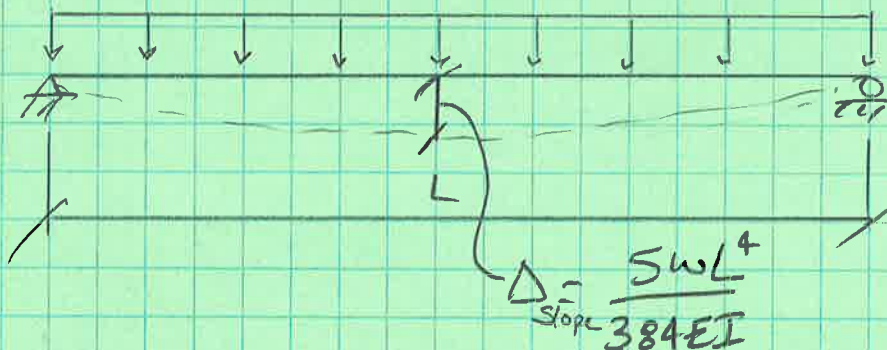
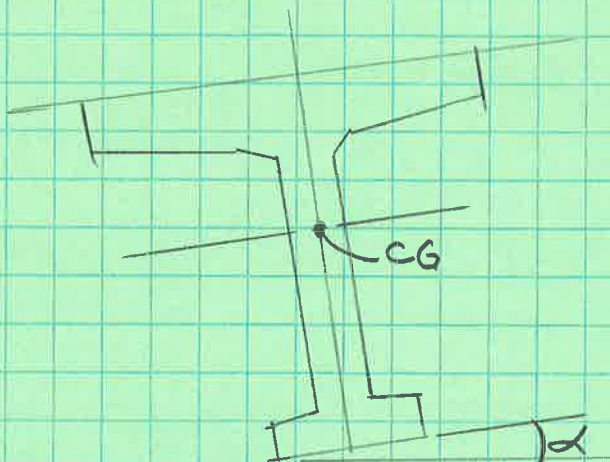


Project			Sheet No. _____ of _____ Sheets	
S.R.	Made By	Check by	Date	Supv

INSTALLATION ON SLOPE

CROWN SLOPE IS NOT BUILT INTO GIRDER
SECTION IS SYMMETRIC

GIRDER INSTALLED SO TOP FLANGE MATCHES CROWN SLOPE



$$w = (A_g \gamma) \sin \alpha$$

- SMALL ANGLES $\sin \alpha \approx \alpha$

$$w = (\alpha A_g \gamma)$$

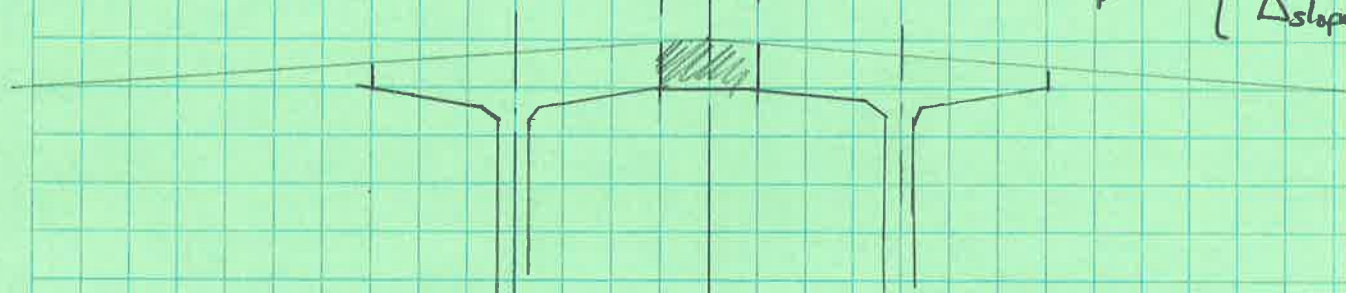


Project			Sheet No.		of		Sheets	
S.R.	Made By	Check by	Date	Supv				

Worst Case Installation AT Crown Pt

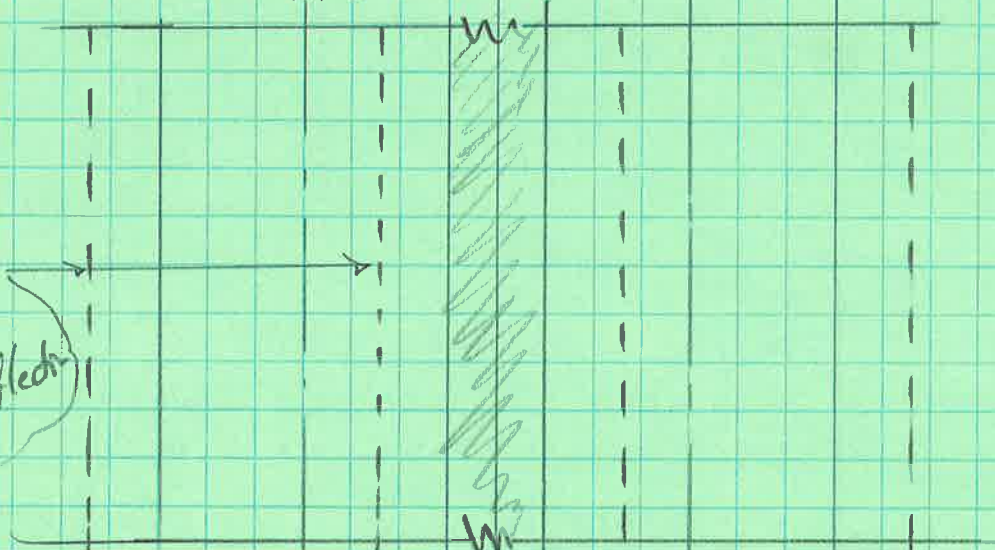
@ MID-SPAN

$$9" + 2\Delta_{\text{sweep}} + 2 \left\{ \begin{array}{l} -\Delta_{\text{ps}} \\ \text{or} \\ \Delta_{\text{slope}} \end{array} \right.$$



GIRDERS TEND TO DEFLECT AWAY FROM EACH OTHER
WIDENING THE JOINT

CGOR JOINT CGDR



Edge of
girder
after
sweep &
lateral deflection

$$9" + 2\Delta_{\text{sweep}} + 2 \left\{ \begin{array}{l} -\Delta_{\text{ps}} \\ \text{or} \\ \Delta_{\text{slope}} \end{array} \right.$$
$$9" + 2(\Delta_{\text{sweep}} + \left\{ \begin{array}{l} -\Delta_{\text{ps}} \\ \text{or} \\ \Delta_{\text{slope}} \end{array} \right.)$$



Project				Sheet No.	of	Sheets
S.R.	Made By	Check by	Date	Supv		

FROM SPAN CAPABILITY TABLES

WF69DG w/ $L = 200$ ft and $W_{tf} = 5$ ft

IS A WORSE CASE (Long girder w/ narrow top flange)

$$\Delta_{\text{SWEEP}} = \left(\frac{1}{8} \right) \left(\frac{200 \text{ ft}}{10 \text{ ft}} \right) = 2.5''$$

$$\gamma = 0.165 \text{ kcf}$$

$$f'_{ci} = 7.5 \text{ ksi}$$

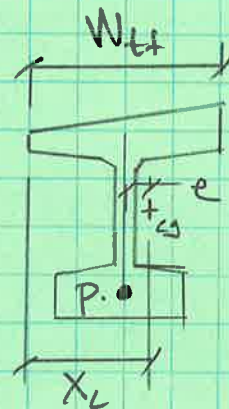
$$E_{ci} = 560 \text{ ksi}$$

$$f'_c = 10 \text{ ksi}$$

$$E_c = 6163 \text{ ksi}$$

$$\text{PRESTRESS AFTER RELEASE} = 2898 \text{ kip}$$

$$\text{eccentricity of PS, } e = X_L - \frac{W_{tf}}{2}$$



$$\Delta_{PS} = \frac{P_e L^2}{8 E_{ci} I}$$

$$\Delta_{\text{slope}} = \frac{5 (\gamma A \gamma) L^4}{384 E_c I}$$



Project S.R.			Sheet No. of Sheets	
Made By	Check by	Date	Supv	

	0%	2%	4%
Area (in ²)	1087.5	1123.5	1159.5
I _y (in ⁴)	150453	161138	171606
X _L (in)	30	30.32	30.62
e (in)	0	0.32	0.62
Δ _{ps} (in)	0	0.74	1.34
Δ _{slope} (in)	0	0.93	1.81

WORST CASE JOINT WIDTH

$$J_1 = 9" + 2(\Delta_{\text{sweep}} + \Delta_{\text{ps}}) = 9 + 2(2.5 + 1.34) = 9 + 7.68 = 16.68" \quad \text{1.32"}$$

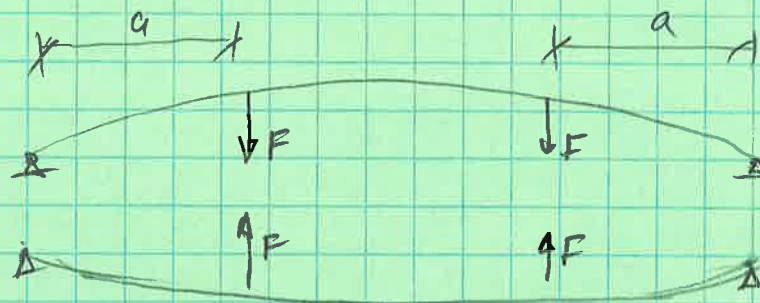
$$J_2 = 9" + 2(\Delta_{\text{sweep}} + \Delta_{\text{slope}}) = 9 + 2(2.5 + 1.81) = 9 + 8.62 = 17.62"$$

@ crown lat. deflection due to prestress tends to close joint (reduce joint width). Sloped girders tend to increase joint width



Project				Sheet No.	of	Sheets
S.R.	Made By	Check by	Date	Supv		

TO ALIGN TOP FLANGES AND BRING THE JOINT WIDTH BACK TO DESIRED VALUE, CONTRACTOR CAN APPLY LATERAL FORCE TO SINK GIRDER TOGETHER



REACTIONS PROVIDED BY GIRDER STOPS

$$\Delta = \frac{Pa}{24EI} (3L^2 - 4a^2)$$

$$a = \frac{L}{3} = 67'$$

$$8.62'' = \frac{P(67)}{24(6163)(171606)} (3 \cdot 200^2 - 4 \cdot 67^2) 12 \frac{\text{in}}{\text{ft}}$$

$$P = 438.6^k \quad \text{IS THIS POSSIBLE?}$$

$$\text{IGNORING LATERAL SWEEP} \quad \Delta = 8.62 - 2(2.5) = 3.62''$$

$$P = 184^k \quad \text{IS THIS POSSIBLE?}$$